

Entropic Measurements of Structure in Graphs

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Abstract

Current information-theoretic measures for network complexity, such as those based on degree distribution or structural symmetry typically yield a single, global value. While insightful, these global metrics fail to distinguish between the internal complexity of a network’s modules and the complexity of the connections between them. Building upon the work of Xiao et al. on symmetry-based structure entropy, we propose a novel hierarchical framework that decomposes a network’s total entropy into two components: a local, intra-module entropy and a global, inter-module entropy. Furthermore, our framework introduces a tunable parameter, α , allowing an analyst to continuously adjust according to their personal definition of complexity between degree-based heterogeneity and symmetry-based structural disorder. This unified model provides a multi-scale and flexible lens for analyzing the architectural principles of complex networks, revealing insights into their modularity, resilience, and evolutionary design.

Motivation

The quantification of complexity in network structures is a cornerstone of modern network science. Measures like degree-based entropy [4] provide a scalable assessment of a network’s heterogeneity. More recently, Xiao et al. [1] introduced a more structurally precise measure, the Entropy based on Automorphism Partition (EAP), which quantifies a network’s disorder based on its underlying symmetries. Such a measure partitions the node set on the basis of equivalence under relabelling, and is a much stronger measure of symmetry than degree. Their work demonstrates that many real-world networks, while appearing moderately heterogeneous from a degree perspective, are almost completely disordered from a structural symmetry perspective (i.e., they possess a high Normalized EAP, or NEAP).

However, these metrics are global. A high global entropy score tells us that the network is complex as a whole, but it does not reveal the source of this complexity. A network could be complex because its constituent parts are internally disordered, or because simple, highly-ordered parts are assembled in a complex and sophisticated manner. This distinction is critical for understanding the design principles, resilience, and function of real-world systems. It can also be seen from Figure 1 and the following discussion that the EAP is not very robust, and one would like a measure that captures the asymmetry predicted by EAP, but is not as dramatic.

This paper introduces a framework to address this gap. We propose a method to decompose a network’s total entropy into components that reflect its modular architecture, and providing a more robust measure of disorder in structure. Our central motivation is to answer the question: *Is a network complex because its building blocks are complex, or because the rules for connecting them are complex?*. In either case, we would want to quantify the complexity.

Formalism

Automorphism Partition Entropy

Definition

Let $G = (V, E)$ be a graph with vertex set V and automorphism group $\text{Aut}(G)$. The automorphism group acts on V , partitioning it into orbits

$$\Pi = \{C_1, C_2, \dots, C_k\},$$

where each C_i is an orbit of vertices that are structurally equivalent under the graph's automorphisms.

The **automorphism partition entropy**[1] is defined as

$$H_{\text{aut}}(G) = - \sum_{i=1}^k p_i \log p_i,$$

where

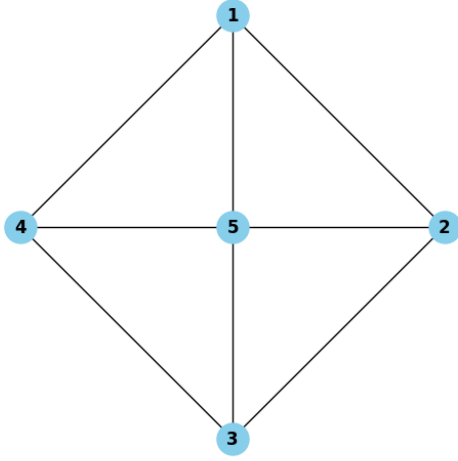
$$p_i = \frac{|C_i|}{|V|}$$

is the fraction of vertices belonging to the i -th orbit.

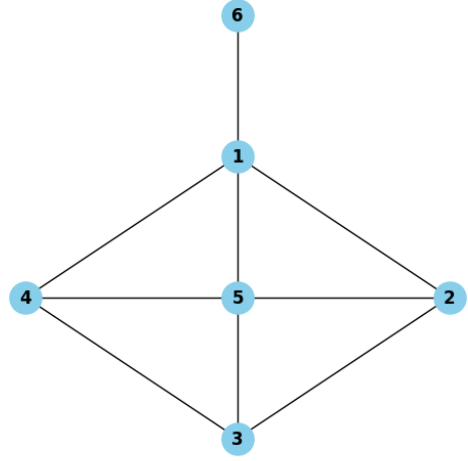
A highly symmetric graph (large orbits) yields low entropy, while an asymmetric graph (mostly singleton orbits) yields high entropy.

Automorphism Partition for Toy Example

Consider a graph $G = (V, E)$ with five vertices: four corner nodes v_1, v_2, v_3, v_4 forming a square, and a central node v_5 connected to all four corners.



(a) Graph G : square with center node.



(b) Graph G' : G with one pendant node.

Figure 1: Toy example graphs used for entropy computation.

The automorphism group $\text{Aut}(G)$ consists of the symmetries of the square that leave the central node fixed. Hence, it includes:

Rotations: $0^\circ, 90^\circ, 180^\circ, 270^\circ$, and Reflections: across both diagonals and both midlines.

Thus, $|\text{Aut}(G)| = 8$, isomorphic to the dihedral group D_4 .

This induces the vertex partition:

$$\Pi = \{C_1, C_2\}, \quad C_1 = \{v_1, v_2, v_3, v_4\}, \quad C_2 = \{v_5\}.$$

Computing the Entropy

The orbit probabilities are

$$p_1 = \frac{4}{5}, \quad p_2 = \frac{1}{5}.$$

Then,

$$H_{\text{aut}}(G) = - \sum_{i=1}^2 p_i \log p_i = - \left(\frac{4}{5} \log \frac{4}{5} + \frac{1}{5} \log \frac{1}{5} \right).$$

Simplified Value

$$H_{\text{aut}}(G) = \log 5 - \frac{4}{5} \log 4.$$

In natural logarithms:

$$H_{\text{aut}}(G) \approx \log 5 - 0.8 \log 4 \approx 1.609 - 0.8(1.386) \approx 0.499.$$

Interpretation

A perfectly asymmetric 5-node graph would have $H_{\text{aut}} = \log 5 \approx 1.609$. Here, $H_{\text{aut}}(G) \approx 0.5$, indicating a moderate degree of symmetry due to the four equivalent corner nodes. Now we shall see how sensitive this measure is to small changes.

Automorphism Partition of G'

Consider modifying the previous 5-node graph by adding one extra vertex v_6 attached (as a pendant/leaf) to corner v_1 . Denote the new graph by G' and its vertex set by $|V'| = 6$.

The leaf on v_1 breaks most square symmetries: any automorphism must fix v_1 (since it is the unique vertex adjacent to a degree-1 node). The only non-trivial automorphism that remains is the reflection across the diagonal through v_1 and v_3 , which swaps v_2 and v_4 while fixing v_1, v_3, v_5, v_6 . Thus $\text{Aut}(G') \cong Z_2$ (size 2), and the orbit partition is

$$\Pi' = \{\{v_2, v_4\}, \{v_1\}, \{v_3\}, \{v_5\}, \{v_6\}\},$$

with orbit sizes 2, 1, 1, 1, 1.

Entropy Computation

Let $p'_i = |C'_i|/6$. The nonzero probabilities are

$$p'_1 = \frac{2}{6}, \quad p'_{2..5} = \frac{1}{6}.$$

Hence,

$$\begin{aligned} H_{\text{aut}}(G') &= - \left(\frac{2}{6} \log \frac{2}{6} + 4 \cdot \frac{1}{6} \log \frac{1}{6} \right) \\ &= - \frac{2}{6} (\log 2 - \log 6) - \frac{4}{6} (-\log 6) \\ &= \log 6 - \frac{2}{6} \log 2. \end{aligned}$$

Numerically (natural logarithms):

$$H_{\text{aut}}(G') \approx 1.5607.$$

Comparison with the 5-node Case

From the previous calculation for G (the 5-node square+center) we had $H_{\text{aut}}(G) \approx 0.5004$. Therefore, the entropy increases by the factor

$$\frac{H_{\text{aut}}(G')}{H_{\text{aut}}(G)} \approx \frac{1.5607}{0.5004} \approx 3.12.$$

So, adding a single pendant node to one corner nearly *triples* the automorphism partition entropy, i.e. a very large relative change caused by a tiny local perturbation.

Why this Shows Lack of Robustness

The automorphism-partition entropy is extremely sensitive to small, local symmetry-breaking changes:

- A single leaf changes orbit structure from $\{4, 1\}$ to $\{2, 1, 1, 1, 1\}$, redistributing orbit probabilities and massively increasing entropy.
- This means H_{aut} can amplify tiny, local modifications into large global entropy shifts which can be problematic if you want a measure stable under small perturbations or noise.

Degree-Based Entropy and Combined Measure

For a network $G = (V, E)$, the degree-based entropy quantifies the heterogeneity of node degrees:

$$H_{\text{deg}}(G) = - \sum_k p_k \log p_k,$$

where p_k is the fraction of nodes with degree k .

To capture both structural symmetry and degree heterogeneity, we construct a combination of the normalized automorphism partition entropy (NEAP) and the normalized entropy of the degree distribution (NEDD):

$$H_{\text{combined}} = \alpha \text{NEAP} + (1 - \alpha) \text{NEDD}, \quad \alpha \in [0, 1].$$

Here, α is a tunable parameter that allows us to continuously interpolate between a purely symmetry-based view ($\alpha = 1$) and a purely degree-based view ($\alpha = 0$).

Unified Hierarchical Entropy Framework

To address the limitations of a single global entropy score, we decompose the network's total complexity into two distinct components: an internal (intra-module) entropy and an inter-module entropy. The total unified entropy is the linear sum of these parts:

$$H_{\text{total}}^{\text{unified}} = H_{\text{internal}}^{\text{unified}} + H_{\text{inter}}^{\text{unified}} \quad (1)$$

Internal Entropy ($H_{\text{internal}}^{\text{unified}}$)

It quantifies the average complexity *within* the network's constituent modules. This measure reveals whether the network's building blocks are internally simple and ordered or complex and heterogeneous.

It is formulated as a weighted sum of the average structural disorder and the average degree heterogeneity of the modules, balanced by the tunable parameter α :

$$H_{\text{internal}}^{\text{unified}} = \alpha \cdot H_{\text{internal}}^{\text{neap}} + (1 - \alpha) \cdot H_{\text{internal}}^{\text{deg}} \quad (2)$$

where:

- $H_{\text{internal}}^{\text{neap}}$ is the weighted average of the NEAP across all modules, reflecting the internal structural complexity.
- $H_{\text{internal}}^{\text{deg}}$ is the weighted average of the NEDD across all modules, capturing the internal degree heterogeneity.

Inter-Module Entropy ($H_{\text{inter}}^{\text{unified}}$)

It quantifies the complexity of the connections *between* modules. This is achieved by analyzing the entropy of the "interface nodes"—the set of nodes that have at least one edge connecting to a different module.

This component measures the diversity of the structural roles and connectivity patterns of these crucial linking nodes:

$$H_{\text{inter}}^{\text{unified}} = \alpha \cdot H_{\text{inter}}^{\text{neap}} + (1 - \alpha) \cdot H_{\text{inter}}^{\text{deg}} \quad (3)$$

where:

- $H_{\text{inter}}^{\text{neap}}$ is the **NEAP** calculated on the partition of the interface nodes, measuring the diversity of their structural roles in bridging modules.
- $H_{\text{inter}}^{\text{deg}}$ is the **NEDD** calculated from the degree distribution of only the interface nodes, measuring the heterogeneity of their connectivity.

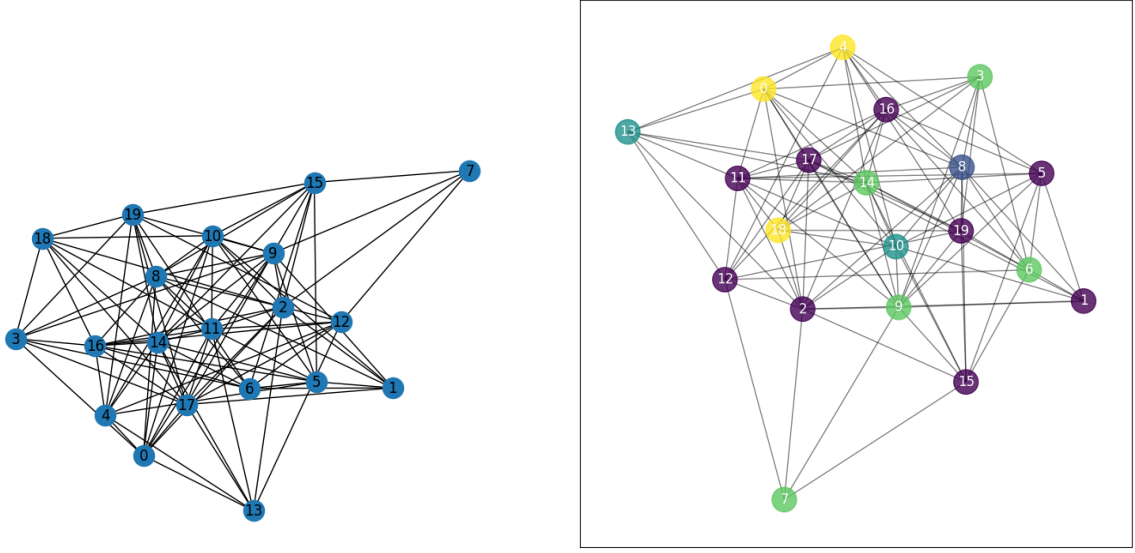


Figure 2: The Erdős-Rényi (ER) random graph $G(n = 20, p = 0.5)$. The left panel shows the generated ER network, and the right panel shows its partition into five random modules.

The respective values for $(H_{\text{internal}}^{\text{unified}})$ and $(H_{\text{inter}}^{\text{unified}})$ for the Erdős-Rényi network used are as follows:

$$H_{\text{internal}}^{\text{unified}} = 0.2515, \quad H_{\text{inter}}^{\text{unified}} = 0.3316, \quad H_{\text{total}}^{\text{unified}} = 0.5830.$$

Hence this means that the entropies are not really additive, and there seems to be some complexity arising from the interconnections. The objective of our approach was to put a number on this, as done above.

Advantages

The proposed framework offers several key advantages over existing global measures:

1. **Multi-Scale Perspective:** It distinguishes between local (intra-module) and global (inter-module) sources of complexity, revealing whether a network is a complex assembly of simple parts or a simple assembly of complex parts.
2. **Tunable Analytical Lens:** The α parameter allows researchers to continuously slide between a computationally fast, degree-based analysis and a structurally precise, symmetry-based analysis. This flexibility can reveal different facets of a network's architecture.
3. **Improved Diagnostic Power:** The model can identify key architectural principles. A network with low H_{internal} but high H_{inter} strongly suggests a modular design with sophisticated integration, a common feature in evolved biological systems and robust engineered systems.
4. **Computational Pragmatism:** By setting $\alpha < 1$, the framework can leverage fast degree-based calculations to remain tractable for very large networks where a full automorphism partition analysis would be computationally infeasible.

Limitations

1. **Dependence on Community Detection:** The accuracy of the framework is contingent on the quality of the initial module or community detection. Different algorithms may yield different partitions, thus affecting the resulting entropy values.

2. **Computational Cost:** For any $\alpha > 0$, the framework requires computing automorphism partitions, which can be computationally intractable for large graphs, especially for the inter-module component.
3. **Theoretical Additivity:** The linear addition $H_{\text{total}} = H_{\text{internal}} + H_{\text{inter}}$ is a theoretical construct. The true relationship between these entropy components may be more complex and is a subject for further research.
4. **Directed Graph Limitation:** There is currently no well-established method to compute the NEAP (Normalized Entropy of Automorphism Partitions) for directed graphs, limiting the applicability of the framework to such networks.

References

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